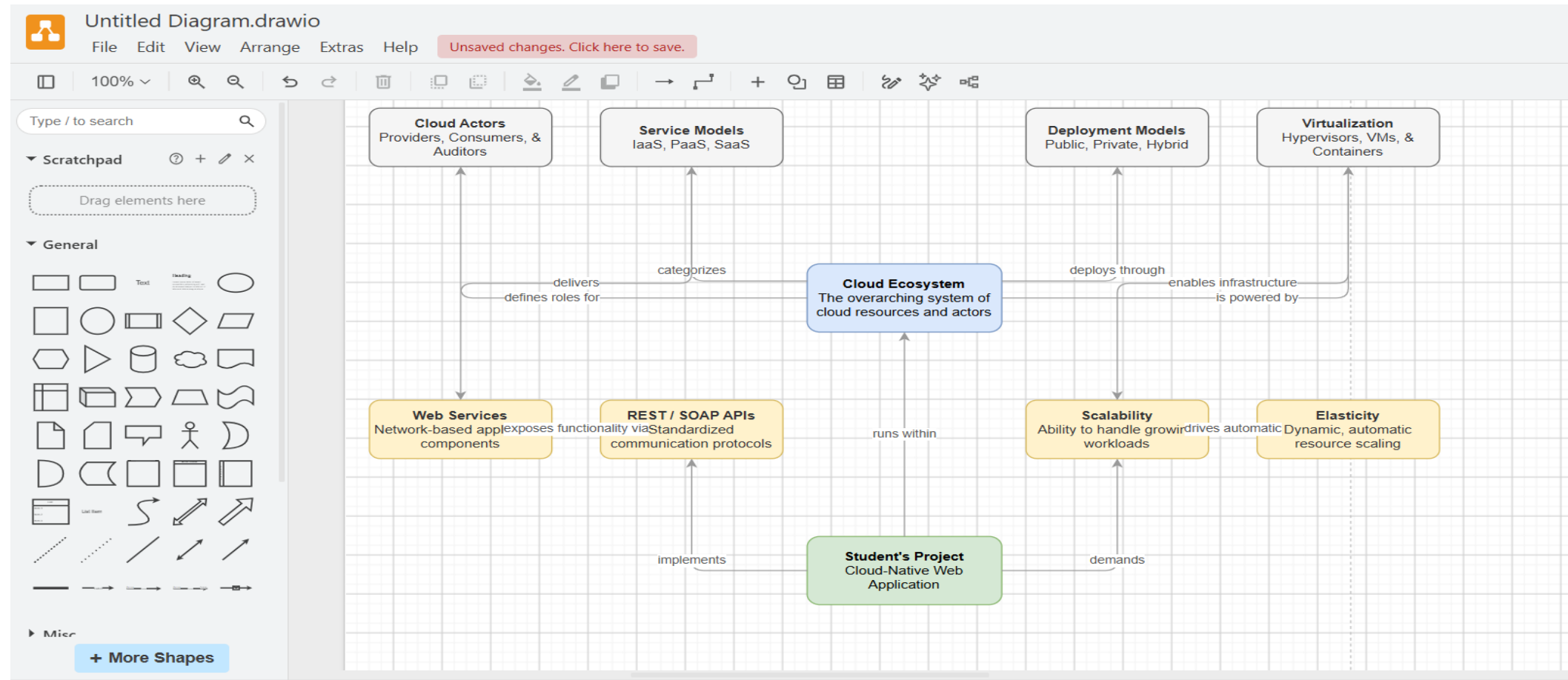


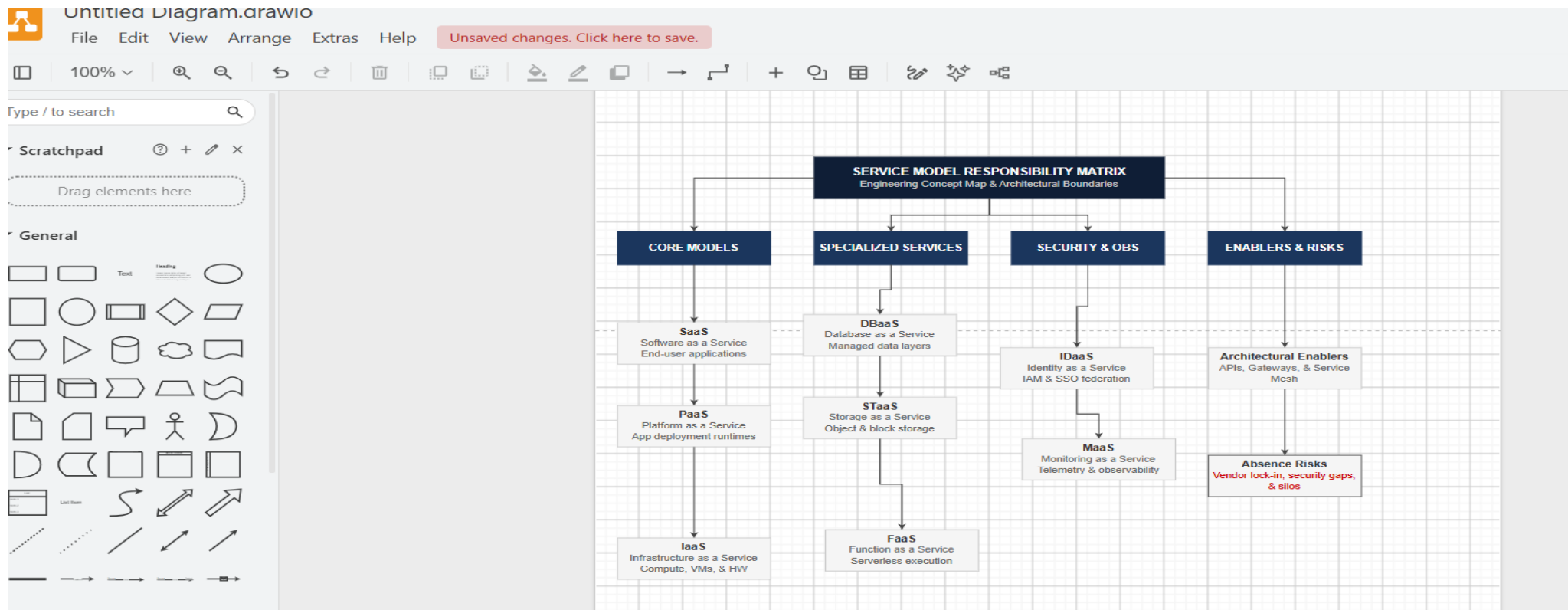
CO1- ASSESSMENT TOOL 1 : Concept Mapping(High)

1. Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why? Create a concept map in draw.io Lucidchart, Miro, Figma, or any suitable diagramming tool. The map must connect cloud ecosystem, cloud actors, service models, deployment models, virtualization, web services, REST/SOAP APIs, scalability, elasticity, and the student's Project.



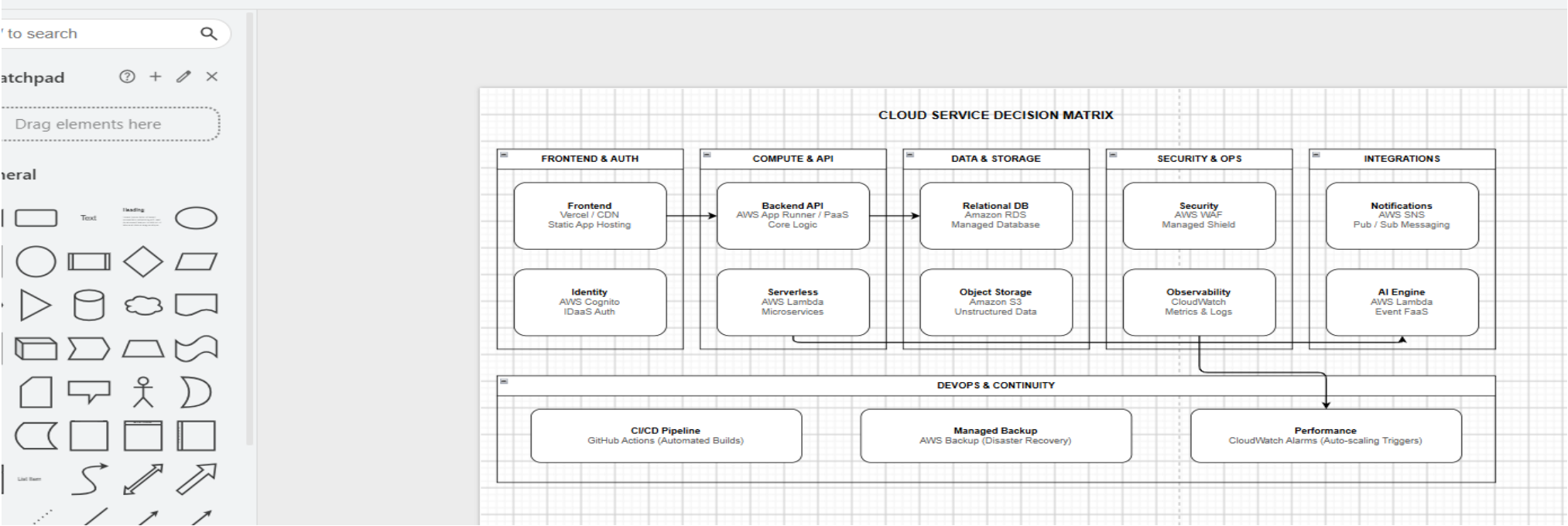
Task 2A: Service Model Responsibility Matrix

Prepare a technical matrix that compares SaaS, PaaS, IaaS, FaaS/serverless, database-as-a-service, storage-as-a-service, identity-as-a-service, and monitoring/logging services. For each selected service category, explain what is managed by the cloud provider, what is managed by the student application team, and why that split fits the Project. Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?

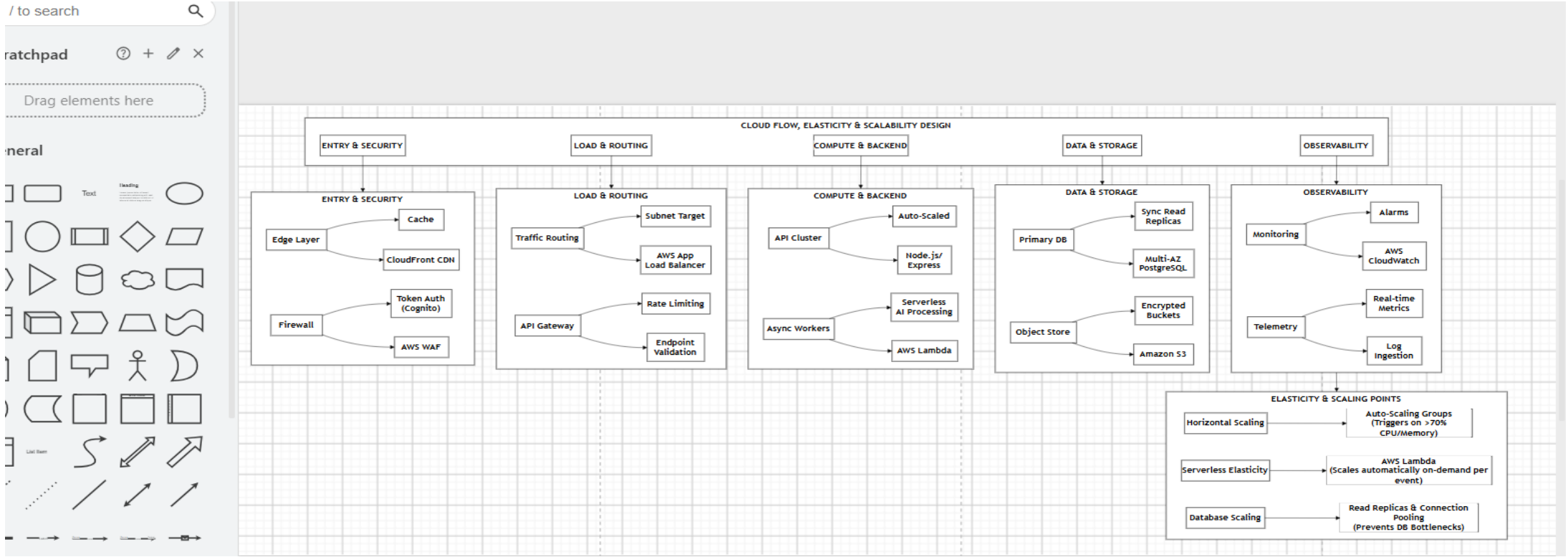


Task 2B: Cloud Service Decision Matrix

Map Project modules to cloud services. Include authentication, frontend hosting, backend/API layer, database, object storage, analytics/logging, notification, AI/Python module, security, CI/CD, backup, and monitoring. Students should justify the most relevant service choices instead of forcing every possible cloud service into the design. Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?



Task 3: API Flow, Elasticity, and Scalability Design Draw a professional cloud-flow diagram for the Project modules. The diagram must show user app or web app, API gateway/backend service, authentication, compute layer, database, object storage, analytics/logging, monitoring, security boundaries, and scaling points. Students may replicate the given visual pattern, but all components and labels must be customized to their own architecture.. Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?



Task 4: Technical Justification

Write a concise justification explaining how the diagram and matrices prove CO1. The justification must discuss service model selection, API role, elasticity, scalability, virtualization or managed compute choice, and cloud responsibility boundaries.. Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigni

